

Dedicated to creating efficient AI solutions by bridging the gap between machine learning and hardware. Skilled in model development, performance optimization, and AI-driven automation.

Education

- 08/2022–05/2027 **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*
(Expected) Advisor: Dr. Abhronil Sengupta
- 08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00* | Thesis: Neuromorphic Computing for Lifelong Learning
- 02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, *CGPA: 3.81/4.00*

Academic Research and Teaching Experience

- 08/2022–Present **Graduate Research Assistant**, *Penn State*, University Park, PA
- Developed advanced ML algorithms including Equilibrium Propagation, Astromorphic Transformer, and RMAAT for long-context sequence modeling, achieving superior accuracy over baseline models.
 - Engineered a hardware-aware training framework integrating model compression (quantization, pruning) for efficient deployment on CIM systems with non-volatile memory.
 - Architecting a high-performance ML accelerator for Transformer/LLM workloads, optimizing dataflow for distributed training and multi-GPU scalability.
- 08/2024–05/2025 **Graduate Teaching Assistant**, *Penn State*, State College, PA
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
 - Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.
- 02/2021–08/2022 **Lecturer**, *University of Liberal Arts Bangladesh (ULAB)*, Dhaka, Bangladesh
- Taught undergraduate courses: Solid State Devices, Digital Circuit Design, Semiconductor Device Physics, Power Electronics.
 - Developed lab modules and supervised projects on semiconductor devices and circuits.

Technical Skills

ML & Data Science	Deep learning, Transformer architectures, LLM fine-tuning and inference, long-context sequence modeling, computer vision, NLP, text/image embeddings, statistical modeling, feature engineering, model development, Neural Architecture Search (NAS), distributed training, multi-GPU, DeepSpeed, AI accelerator development, Systolic Arrays, HPC
ML Tools	Hugging Face Transformers, ONNX, TensorRT, PyTorch, TensorFlow, JAX, scikit-learn, Keras, Matplotlib, Seaborn, Pandas, NumPy, Jupyter, Weights & Biases
Hardware & EDA	GPU programming, Cadence Virtuoso, Spectre, HSPICE, TCAD, COMSOL, ModelSim, Synopsys (Design Compiler, PrimeTime, VCS), Quantization (PTQ/QAT), pruning, distillation, mixed-precision (FP16/INT8), hardware-aware training, FPGA prototyping, CIM/PIM, near-memory computing
Programming	Python, CUDA, C/C++, MATLAB, Verilog, Shell/Bash, Linux/Unix, Docker, Git
DTCO & STCO	Design-technology and system-technology co-optimization, cross-layer PPA, process-aware design, system-architecture codesign
AI Dev.	Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development
Collaboration	Git, Microsoft Office, Google Workspace

Industrial Experience

- 05/2025–**Graduate Technical Intern, Intel Corporation, Hillsboro, OR**
07/2025
 - Developed AI-driven predictive models for process yield optimization and critical dimension (CD) control in semiconductor manufacturing.
 - Implemented machine learning algorithms to predict optimal deposition parameters and automate recipe generation.
 - Created comparative analysis framework between AI-based models and statistical models (JMP) for enhanced prediction accuracy.
- 09/2020–**R&D Engineer, SEMWAVES Ltd., London, UK, part-time**
07/2021
 - Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
 - Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Research Projects

- Astromorphic Transformer** Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture leveraging neuron-astrocyte interactions to emulate self-attention mechanisms. Incorporated Hebbian and presynaptic plasticities with non-linearities and feedback, achieving superior accuracy and faster convergence on sentiment classification (IMDB), image classification (CIFAR-10), and language modeling (WikiText-2) tasks with improved perplexity compared to conventional transformers. See publication: [IEEE TCDS 2025].
- Neuromorphic Cybersecurity with Lifelong Learning** Lead Student Researcher, 2023–2025. Developed a Hierarchical Dynamic Spiking Neural Network (D-SNN) for Network Intrusion Detection Systems combining static SNN detection (94.3% accuracy) with adaptive dynamic SNN classification using GWR-inspired structural plasticity and novel Adaptive STDP (Ad-STDP) learning. Achieved 85.3% overall accuracy on UNSW-NB15 benchmark in semi-supervised lifelong learning, demonstrating superior adaptation to new attack types while mitigating catastrophic forgetting (5.3% improvement over baseline). See publication: [arXiv], [ICONS 2025].
- DCNN-LSTM Audio Classification** Co-Author, 2022. Developed a DCNN-LSTM based audio classification system combining multiple feature engineering and data augmentation techniques. Achieved robust performance on diverse audio datasets. See publication: [ICOL 2021, Springer].

Select Publications

- [1] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. “RMAAT: Astrocyte-Inspired Memory Compression and Replay for Efficient Long-Context Transformers”. In: *International Conference on Learning Representations (ICLR)*. 2026. URL: <https://openreview.net/forum?id=sTkJdbVxsI>.
- [2] **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. “Trilinear Compute-in-Memory Architecture for Energy-Efficient Transformer Acceleration”. In: *arXiv preprint arXiv:2604.07628* (2026).
- [3] **Md Zesun Ahmed Mia**, Malyaban Bal, Sen Lu, George M. Nishibuchi, Suhas Chelian, Srinivasan, and Abhronil Sengupta. “Neuromorphic Cybersecurity with Semi-Supervised Lifelong Learning”. In: *Proceedings of the International Conference on Neuromorphic Systems (ICONS '25)*. IEEE Press, 2025, pp. 235–238. DOI: 10.1109/ICONS69015.2025.00043.
- [4] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. “Delving deeper into astromorphic transformers”. In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).

Recognitions

- Melvin P. Bloom Memorial Outstanding Doctoral Research Award (2026)
- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024), Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)
- Student Member, IEEE (2015-Present), Executive Member, EDS, IEEE BD (2021-2022)